

## VEREENIGING, South Africa

WMO: 683530

Lat: 26.570S

Lon: 27.959E

Elev: 4859

StdP: 12.29

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 28.7       | 31.7 | 8.5                           | 10.2 | 55.1 | 12.0 | 12.1 | 54.9 | 19.6                  | 58.4 | 17.5 | 60.4 | 4.1                   | 330  | 0.392 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 12            | 21.4                   | 88.2            | 58.9 | 86.3 | 59.0 | 84.5 | 59.1 | 66.3                | 76.8 | 65.5 | 76.0 | 64.7 | 75.3 | 8.2                  | 310  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.0                            | 103.3 | 68.9 | 62.6 | 101.7 | 68.6 | 61.2 | 96.8 | 68.2 | 33.9          | 77.6 | 33.3 | 76.4 | 32.6 | 75.9 | 73.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 18.6          | 16.0 | 13.9 | DB                         | 23.0 | 92.9               | 3.1 | 3.6                                                | 20.8 | 95.5       | 18.9 | 97.6       | 17.2 | 99.7       | 14.9 | 102.3 | (4) |
| (5)               |      |      | WB                         | 19.8 | 68.5               | 3.2 | 2.3                                                | 17.5 | 70.2       | 15.6 | 71.5       | 13.9 | 72.8       | 11.6 | 74.4  | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 62.1  | 69.4  | 69.0  | 67.0  | 61.2  | 55.1  | 50.2  | 49.9  | 55.2  | 62.7  | 67.3  | 68.2  | 70.3 | (6)  |
| (7)  |                                                                                  | DBStd       | 8.90  | 5.80  | 4.88  | 4.49  | 4.81  | 4.65  | 4.30  | 4.52  | 6.12  | 6.15  | 5.32  | 4.77  | 3.91 | (7)  |
| (8)  |                                                                                  | HDD50       | 139   | 0     | 0     | 0     | 2     | 12    | 46    | 55    | 22    | 2     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD65       | 1917  | 32    | 22    | 29    | 128   | 308   | 445   | 468   | 307   | 110   | 38    | 23    | 7    | (9)  |
| (10) |                                                                                  | CDD50       | 4553  | 600   | 533   | 528   | 339   | 170   | 52    | 52    | 183   | 384   | 536   | 548   | 628  | (10) |
| (11) |                                                                                  | CDD65       | 853   | 166   | 135   | 92    | 15    | 1     | 0     | 0     | 3     | 42    | 108   | 121   | 170  | (11) |
| (12) |                                                                                  | CDH74       | 8275  | 1318  | 1077  | 801   | 258   | 64    | 3     | 5     | 183   | 821   | 1234  | 1113  | 1398 | (12) |
| (13) |                                                                                  | CDH80       | 2126  | 364   | 264   | 158   | 28    | 3     | 0     | 0     | 13    | 205   | 373   | 313   | 405  | (13) |
| (14) | Wind                                                                             | WSAvg       | 5.6   | 5.4   | 4.7   | 4.5   | 4.2   | 4.6   | 4.8   | 4.9   | 6.2   | 6.8   | 7.6   | 7.3   | 6.5  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 24.9  | 4.6   | 3.1   | 3.0   | 1.7   | 0.6   | 0.2   | 0.1   | 0.3   | 0.8   | 2.6   | 3.5   | 4.8  | (15) |
| (16) |                                                                                  | PrecMax     | 35.7  | 10.2  | 6.6   | 8.1   | 4.9   | 3.8   | 1.0   | 1.5   | 1.5   | 3.7   | 8.6   | 8.0   | 8.3  | (16) |
| (17) |                                                                                  | PrecMin     | 14.2  | 1.6   | 0.9   | 0.5   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.7   | 0.3   | 1.9  | (17) |
| (18) |                                                                                  | PrecStd     | 5.7   | 2.0   | 1.6   | 1.6   | 1.3   | 0.9   | 0.2   | 0.3   | 0.4   | 0.9   | 1.8   | 1.7   | 1.7  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 91.2  | 89.3  | 87.4  | 83.2  | 79.9  | 73.7  | 73.7  | 81.9  | 87.7  | 90.2  | 91.1  | 91.1 | (19) |
| (20) |                                                                                  |             | MCWB  | 61.9  | 60.9  | 59.3  | 57.8  | 53.8  | 50.3  | 48.9  | 54.0  | 55.2  | 57.2  | 58.4  | 62.0 | (20) |
| (21) |                                                                                  | 2%          | DB    | 86.3  | 86.0  | 84.1  | 80.3  | 75.4  | 70.5  | 71.3  | 78.8  | 84.6  | 87.5  | 86.3  | 86.4 | (21) |
| (22) |                                                                                  |             | MCWB  | 61.6  | 61.2  | 59.3  | 56.9  | 52.4  | 48.7  | 48.0  | 52.8  | 54.1  | 57.2  | 58.2  | 60.7 | (22) |
| (23) |                                                                                  | 5%          | DB    | 84.1  | 83.0  | 81.2  | 77.1  | 73.2  | 68.1  | 68.4  | 75.5  | 82.4  | 84.3  | 84.0  | 84.3 | (23) |
| (24) |                                                                                  |             | MCWB  | 61.9  | 61.7  | 59.6  | 57.1  | 51.7  | 48.0  | 47.1  | 50.9  | 53.8  | 56.5  | 58.6  | 61.0 | (24) |
| (25) |                                                                                  | 10%         | DB    | 81.0  | 80.7  | 78.9  | 74.5  | 70.0  | 65.9  | 66.0  | 72.0  | 79.0  | 81.0  | 80.6  | 82.0 | (25) |
| (26) |                                                                                  |             | MCWB  | 62.8  | 62.4  | 59.5  | 56.5  | 50.9  | 47.1  | 46.5  | 49.5  | 53.2  | 56.7  | 59.0  | 61.0 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 67.4  | 67.4  | 65.5  | 63.6  | 58.3  | 52.9  | 51.7  | 57.1  | 58.8  | 63.0  | 64.8  | 67.2 | (27) |
| (28) |                                                                                  |             | MCDWB | 78.1  | 78.3  | 75.7  | 72.5  | 70.2  | 67.2  | 67.8  | 73.9  | 78.0  | 77.6  | 77.1  | 78.4 | (28) |
| (29) |                                                                                  | 2%          | WB    | 66.4  | 66.1  | 64.3  | 61.8  | 55.7  | 51.1  | 50.1  | 54.9  | 57.2  | 61.3  | 63.5  | 65.5 | (29) |
| (30) |                                                                                  |             | MCDWB | 76.7  | 76.1  | 73.7  | 70.9  | 67.9  | 65.5  | 66.7  | 74.5  | 76.4  | 76.4  | 76.3  | 77.5 | (30) |
| (31) |                                                                                  | 5%          | WB    | 65.5  | 65.2  | 63.2  | 60.2  | 54.2  | 49.8  | 48.7  | 53.2  | 55.8  | 60.2  | 62.4  | 64.6 | (31) |
| (32) |                                                                                  |             | MCDWB | 75.6  | 75.5  | 72.0  | 69.8  | 67.2  | 63.8  | 65.1  | 71.1  | 75.3  | 75.1  | 75.2  | 76.7 | (32) |
| (33) |                                                                                  | 10%         | WB    | 64.7  | 64.3  | 62.0  | 58.8  | 52.7  | 48.3  | 47.4  | 51.3  | 54.6  | 59.1  | 61.5  | 63.5 | (33) |
| (34) |                                                                                  |             | MCDWB | 74.6  | 75.0  | 71.1  | 68.8  | 66.2  | 62.1  | 63.7  | 68.3  | 74.6  | 73.4  | 73.9  | 75.3 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 22.2  | 23.1  | 22.9  | 25.3  | 29.6  | 31.1  | 31.4  | 31.5  | 29.1  | 25.5  | 22.6  | 21.4 | (35) |
| (36) |                                                                                  |             | MCDBR | 26.5  | 27.4  | 27.0  | 29.3  | 33.6  | 34.6  | 35.2  | 34.1  | 33.0  | 30.0  | 27.5  | 26.0 | (36) |
| (37) |                                                                                  |             | MCWBR | 7.9   | 8.9   | 8.9   | 11.3  | 14.6  | 16.4  | 16.3  | 13.9  | 12.6  | 10.5  | 9.4   | 8.1  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 19.3  | 20.4  | 19.9  | 22.5  | 26.8  | 29.8  | 31.1  | 30.4  | 28.8  | 24.6  | 22.4  | 21.1 | (38) |
| (39) |                                                                                  |             | MCWBR | 7.2   | 7.7   | 7.8   | 10.2  | 12.4  | 14.8  | 15.1  | 13.8  | 11.1  | 9.7   | 9.0   | 8.1  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.395 | 0.397 | 0.382 | 0.332 | 0.295 | 0.278 | 0.280 | 0.335 | 0.377 | 0.412 | 0.369 | 0.378 | (40) |      |
| (41) |                                                                                  | taud        | 2.303 | 2.301 | 2.319 | 2.428 | 2.509 | 2.527 | 2.508 | 2.350 | 2.244 | 2.185 | 2.344 | 2.339 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 301   | 295   | 290   | 290   | 289   | 288   | 292   | 285   | 286   | 287   | 307   | 306   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 45    | 43    | 41    | 33    | 28    | 26    | 27    | 35    | 43    | 48    | 43    | 43    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2107  | 1977  | 1754  | 1499  | 1360  | 1235  | 1342  | 1587  | 1903  | 2033  | 2148  | 2175  | (44) |      |
| (45) |                                                                                  | RadStd      | 231   | 175   | 154   | 129   | 70    | 56    | 37    | 76    | 87    | 145   | 183   | 132   | 45   |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (13 neighbors) |  | +1.17 | +1.33   | N/A    | N/A     | -0.85 | -0.94 | N/A         | -326  | +421  | +157  |

Nomenclature: See separate page